UNIGUARD AI

AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Smart India Hackathon (SIH26145) • TEAM LIQUID

1. Executive Summary & Problem Context

Critical national infrastructure (power grids, defense telemetry, nuclear enclaves, banking ledgers, SCADA systems) relies heavily on strict network isolation. Traditional Network Intrusion Detection Systems (NIDS) and host monitoring agents depend fundamentally on **bidirectional communication**: they track stateful TCP handshakes, query live endpoints, and inject active probes. In strictly air-gapped and unidirectional optical diode networks, return communication is physically prohibited. Standard security tools fail completely in these environments, creating an acute operational blind spot.

2. The UNIGUARD AI Solution

UNIGUARD AI is a passive, high-throughput cyber threat intelligence platform engineered to analyze strictly unidirectional network traffic. Operating with **zero active probing, zero return-path dependency, and 100% passive listening**, UNIGUARD AI ingests raw packet streams, reconstructs single-direction 5-tuple flows, extracts 27 deterministic temporal and statistical features, and executes a two-tier machine learning pipeline to classify threats and detect zero-day anomalies in real time.

RAW PCAP/SPAN	PASSIVE INGEST	5-TUPLE AGGREGATION	27 FEATURES	TWO-TIER ML	SHA-256 LEDGER
Physical/SPAN mirror	100% Read-Only (0% TX)	Directional sliding window	Entropy & Dynamics	4-Class + IsolationForest	Immutable Evidence

3. Key Architectural Innovations

• **Zero-Emission Passive Telemetry:** Receives packets on RX line only; never transmits packets into the monitored network.

• **27 Deterministic Flow Features:** Computes packet-size Shannon entropy, inter-arrival jitter, and flag ratios in sub-millisecond time.

• **Two-Tier ML Core:** Combines supervised HistGradientBoosting (4-class) with unsupervised Isolation Forest for zero-day outliers.

• **Cryptographic Evidence Integrity:** Automatic SHA-256 block hashing guarantees non-repudiation for incident response.

PAGE 2: SYSTEM ARCHITECTURE & WORKFLOW

1. High-Level Architecture Flow

Architecture Stage	Component Role	Data Produced	Isolation Guarantee
1. Monitored Network	Source operational VPC / servers	Raw IP packets	Target under passive surveillance
2. Passive TAP / SPAN	Optical beam splitter / switch mirror	Read-only packet clones	No inline latency or packet mutation
3. Data-Diode Boundary	Physical/logical one-way barrier	Unidirectional stream	RX enabled; return TX physically blocked
4. Flow Aggregator	Scapy / DPKT sliding-window engine	5-tuple directional flows	No reciprocal handshake requirement
5. 27-Feature Engine	Statistical & temporal calculator	Normalized feature vector	Deterministic math across single flows
6. Two-Tier ML Core	HistGradientBoosting + IsolationForest	Threat class & anomaly score	Zero-day outlier & calibrated probability
7. Evidence Ledger	SHA-256 cryptographic chain	Immutable audit block	Tamper-evident forensic ledger
8. FastAPI & UI	Uvicorn REST + React 19 dashboard	Visual SOC analytics	Real-time telemetry & PDF/JSON exports

2. Operational Workflow Breakdown

Step 1 — Ingestion: Mirrored Ethernet frames pass through the unidirectional boundary into the aggregator.
Step 2 — Reconstruction: Packets sharing (Src IP, Dst IP, Src Port, Dst Port, Protocol) are aggregated into discrete temporal flows.
Step 3 — Feature Extraction: The engine extracts volume metrics, packet-size Shannon entropy, inter-arrival time (IAT) variance, and TCP flag states.
Step 4 — Two-Tier Classification: Tier 1 evaluates calibrated multi-class decision trees (Benign, DDoS, Recon, C2). Tier 2 evaluates isolation path lengths.
Step 5 — Forensic Sealing: Feature vectors, timing metadata, and model outputs are hashed with SHA-256 and committed to the evidence ledger.

3. Software-Only Prototype Distinction

UNIGUARD AI is implemented as a **complete, self-contained software-only prototype**. The data diode boundary is modeled in software to reflect real-world unidirectional optical diode characteristics. Physical optical diode hardware integration represents an optional, modular future deployment layer.

PAGE 3: TECHNOLOGY STACK & KEY DEFINITIONS

1. Technology Stack Implementation Matrix

Layer	Technology	Version / Library	Operational Function
Frontend UI	React, TypeScript, Vite	React 19, TS 6.0, Vite 8	High-performance SPA with 3D isometric visualizations
Styling & Design	Tailwind CSS, Lucide	Tailwind 4, Lucide 1.34	Responsive enterprise cybersecurity theme (dark/light)
Backend API	Python, FastAPI, Uvicorn	FastAPI 0.110, Python 3.14	Asynchronous microservice for REST and streaming PCAP
Network Analysis	Scapy, DPKT	Scapy 2.5.0, DPKT 1.9.8	Binary PCAP/PCAPNG parsing & packet header extraction
Machine Learning	scikit-learn, NumPy, Pandas	scikit-learn 1.4+, Joblib	HistGradientBoosting, IsolationForest, RobustScaler
Database	Supabase, PostgreSQL	Cloud PostgreSQL 15	Relational incident storage & SHA-256 evidence ledger
Security Hardening	Custom Starlette Middlewares	Python hashlib, BaseHTTP	Rate limiter, magic-byte checks, path traversal bounds

2. Key Terms & Acronym Reference

Term	Full Form	Context in UNIGUARD AI
AI / ML	Artificial Intelligence / Machine Learning	Algorithmic decision-making and statistical threat classification
IP / TCP / UDP	Internet Protocol / TCP / UDP	Network and transport layers providing 5-tuple flow identity
PCAP / PCAPNG	Packet Capture / Next Generation	Standard binary formats ingested for authentic traffic evaluation
DDoS	Distributed Denial of Service	Volumetric attack detected via high packets/sec and SYN ratios
C2	Command and Control	Covert botnet beaconing detected via low IAT jitter and periodicity
DGA / DNS	Domain Generation Algorithm / DNS	Algorithmic malware domains detected via Shannon character entropy
TLS / HTTPS	Transport Layer Security / HTTP Secure	Encrypted sessions analyzed via metadata without payload decryption
REST API / JSON	Representational State Transfer / JSON	Stateless client-server telemetry exchange protocol
SHA-256	Secure Hash Algorithm 256-bit	Cryptographic one-way hash securing evidence non-repudiation
CORS / RLS	Cross-Origin Resource Sharing / RLS	Browser origin boundaries and PostgreSQL Row Level Security

PAGE 4: 27-FEATURE ENGINE & MACHINE LEARNING ARCHITECTURE

1. Canonical 27-Feature Extraction Matrix

Category	Feature Name	Mathematical / Logical Definition	Threat Relevance
Volume & Time	duration_seconds packet_count byte_count packets_per_second bytes_per_second	Active flow lifetime (sec) Total packet tally in slice Total byte payload tally Packet density: N/duration Byte density: Bytes/duration	DDoS floods, exfiltration bursts, stealth scans
Packet Size Entropy	mean_packet_size std_packet_size min_packet_size max_packet_size packet_size_entropy	Mean size of packets Standard deviation in size Minimum packet magnitude Maximum packet magnitude Shannon Entropy: -sum(p*log2(p))	Detects structured payloads, tunneling, and fixed-size beaconing
IAT Dynamics	mean_iat_seconds std_iat_seconds min_iat_seconds max_iat_seconds iat_jitter_seconds burstiness_ratio	Mean packet inter-arrival gap IAT standard deviation Smallest inter-packet gap Largest idle time gap First-order IAT variance Burst packets / total packets	C2 periodic beaconing exhibits extremely low IAT jitter
TCP Flags	syn_count, ack_count fin_count, rst_count psh_count, urg_count syn_ratio, ack_ratio, rst_ratio	Raw TCP control flag counts and proportional flag distributions across unidirectional flow frames	SYN floods, RST connection tears, stealth scanning flags
Port Indicators	is_well_known_dst_port is_ephemeral_src_port	Dst Port in 0..1023 (Boolean) Src Port >= 1024 (Boolean)	Identifies targeted standard services vs random ephemeral ports

2. Two-Tier AI Inference Engine

Tier 1 — HistGradientBoostingClassifier: Multi-class gradient boosted decision trees bin continuous features into 256 histograms, yielding ultra-fast sub-millisecond evaluations with built-in resilience to feature scaling and missing values. Trained on class-balanced captures to classify: BENIGN, DDOS, RECON_SCAN, and C2_BEACONING.

Tier 2 — Isolation Forest (Novelty/Anomaly Core): Evaluates the average path length required to isolate a flow across an ensemble of 100 random trees. Flows with shorter tree paths receive anomaly scores ≤ 0.0 and are flagged as statistical outliers, providing a critical safety net for zero-day attacks.

PAGE 5: THREAT DETECTION & REAL-WORLD PCAP VALIDATION

1. Six-Category Threat Detection Matrix

Threat Category	Detection Engine	Underlying Behavioral Indicators	Verification Status
1. Volumetric / DDoS	Tier 1 Supervised ML	High packets_per_second, high syn_ratio, low entropy	VERIFIED (Authentic PCAP)
2. Botnet C2 Beaconing	Tier 1 Supervised ML	Low iat_jitter_seconds, fixed periodicity, burstiness	VERIFIED (CTU-13 Neris)
3. Recon / Port Scan	Tier 1 Supervised ML	Dispersed dst_port, high rst_ratio, small packet sizes	VERIFIED (Nmap PCAP)
4. DGA / DNS Tunneling	Metadata & Entropy	High Shannon character entropy in subdomains	VERIFIED (Metadata Engine)
5. TLS Encrypted Malware	Metadata & Jitter	Record size distributions, TLS cipher negotiation	VERIFIED (Metadata Engine)
6. Data Exfiltration	Statistical Heuristic	High byte-to-duration ratios, anomalous upload spikes	HEURISTIC RISK ONLY

2. Authentic Multi-Capture PCAP Benchmark

Capture Dataset	Attack Scenario	Evaluated Flows	Primary Detections	Accuracy / Recall
CSE-CIC-IDS2018 (Wed14)	LOIC HTTP Flooding (DDoS)	1,240 Flows	DDOS (Tier 1 Hits)	100.0% DDoS Recall
CTU-13 Dataset (Capture)	Neris IRC C2 Beaconing	420 Flows	C2_BEACONING	98.8% C2 Recall
CIC-IDS2017 (Fri02)	Ares Botnet Command Channel	310 Flows	C2 & Anomaly Outlier	97.4% Botnet Recall
Internal Recon Capture	Nmap SYN Stealth Port Scan	288 Flows	RECON_SCAN (Tier 1)	96.2% Recon Recall

3. Unseen Test Set Performance Verification

- **Overall Multi-Class Accuracy:** 82.24% (Evaluated on completely unseen test captures with 0 split leakage)
- **Weighted F1-Score:** 0.8700 | **Benign False Positive Rate (FPR):** 2.82%
- **Mean Inference Latency:** 14.25 ms/flow (p95: 16.22 ms, p99: 22.33 ms) | **Throughput:** 70.2 flows/sec

PAGE 6: SECURITY FEATURES, DATA SAFETY & EVIDENCE

1. Defensive Security Hardening Matrix

Security Layer	Mechanism Implemented	Defensive Purpose
IP Rate Limiting	InMemoryRateLimiter sliding-window limiter	Prevents API denial-of-service (30 uploads/min limit)
Payload Bounds	50 MB MAX_UPLOAD_SIZE_BYTES limit	Protects backend memory from exhaustion via giant captures
Magic-Byte Check	4-byte header signature validator	Rejects scripts/binaries disguised as PCAP/PCAPNG
Path Traversal Defense	sanitize_safe_filename() validator	Rejects '..', '/', '\' and non-whitelisted extensions
HTTP Security Headers	SecurityHeadersMiddleware (Custom)	X-Content-Type-Options: nosniff, X-Frame-Options: DENY
Origin Protection	Configurable CORS (CORS_ORIGINS)	Restricts browser API access to authorized frontend origins
Secret Isolation	.env file configuration with Git ignore	Prevents database service-role key leakage to repositories

2. SHA-256 Forensic Evidence Chaining

Every detected threat incident is cryptographically sealed into the forensic_evidence_ledger table. The SHA-256 hash is computed across the incident UUID, flow 5-tuple, feature vector, and the preceding block's hash:
Block_Hash = SHA256(Incident_ID || 5-Tuple || Entropy || Features || Previous_Block_Hash)
This chained structure guarantees that any historical alteration of forensic records invalidates downstream hashes, enabling tamper-evident audit trails for cybersecurity compliance.

3. Data Safety & Confidentiality Principles

- **No Active Network Emission:** UNIGUARD AI runs in pure passive mode; zero packets are sent to monitored networks.
- **Zero Decryption Exposure:** Inspection operates strictly over outer packet headers, timing, and entropy.
- **Ephemeral Capture Handling:** Uploaded PCAP slices are processed in isolated temporary storage and cleared immediately.

PAGE 7: DATABASE SCHEMA & API ARCHITECTURE

1. Supabase Relational Database Schema (ER)

Table Name	Primary Key	Key Attributes & Types	Foreign Keys / Relations
system_audit_logs	id (UUID)	event_type, component, severity, message, metadata (JSONB)	Standalone audit table
diode_configurations	id (UUID)	channel_name (UNIQUE), mode, is_active, max_throughput_mbps	Hardware state registry
detection_models	id (UUID)	model_id (UNIQUE), name, version, algorithm, accuracy, is_active	Active ML metadata registry
threat_incidents	id (UUID)	incident_code (UNIQUE), threat_type, severity, confidence, IPs	1-to-Many with evidence ledger
forensic_evidence_ledger	id (UUID)	incident_id (FK), sha256_hash (UNIQUE), previous_block_hash	FK -> threat_incidents(id) CASCADE

2. FastAPI REST & Streaming Endpoint Matrix

HTTP	Endpoint Route	Parameters	Function & Response Payload
GET	/api/health	None	Returns API status, active ML phase, Supabase connectivity
GET	/api/db/status	None	Verifies live PostgreSQL database latency and tables
GET	/api/ml/model-info	None	Returns active Phase 3I model metadata, 27 features, metrics
GET	/api/ml/samples	None	Lists verified sample captures (LOIC DDoS, Ares Botnet)
POST	/api/analyze/pcap	file (Binary), max_flows	Full PCAP upload analysis, 5-tuple extraction, 4-class ML
POST	/api/analyze/sample-pcap	filename, max_flows	Pre-loaded capture evaluation with timing breakdown
GET	/api/stream/sample-pcap	filename, chunk_size	Incremental streaming telemetry feed for real-time UI

PAGE 8: UI/UX APPLICATION MODULES & REPORTING

1. Frontend Dashboard Modules (React 19 SPA)

Module / Page	Route	Key Functional Capabilities
1. Overview	/	3D isometric architecture visualization, quick PCAP eval, pipeline status, KPI cards
2. Live Traffic	/live-traffic	Real-time flow table, packet rates, directional arrows, anomaly score pills
3. Threats	/threats	Prioritized incident queue, severity badges (Critical/High/Med), attack breakdowns
4. Evidence	/evidence	Cryptographic ledger viewer, SHA-256 block hash inspector, tamper-evident records
5. Models	/models	Phase 3I model registry, 27 feature rankings, confusion matrix, accuracy metrics
6. Analytics	/analytics	Temporal entropy charts, protocol distributions, volume density visualizers
7. Reports	/reports	Printable executive PDF dossier, machine-readable JSON export, Markdown summary
8. System Health	Modal	Subsystem telemetry modal (optical diode link state, DB status, API engine)
9. Team Profile	Modal	Interactive Team Liquid roster modal with member contact cards and clipboard copy

2. Team Liquid Engineering Roster

Member Name	Designation	Registration No.	Department / Institution	Contact
ANGU HARI KARTHICK M	Team Lead	25105018	B.E. CSE, R-Smart Intellect Eng.	8072522246
SRI POOJITH RUTHRAN	Core Member	25105225	B.E. CSE, R-Smart Intellect Eng.	8248082282
HARINISHA S	Core Member	25105085	B.E. CSE, R-Smart Intellect Eng.	8667751810

PAGE 9: TESTING, PERFORMANCE & DIFFERENTIATION

1. Verification & Automated Test Suite Results

Test Suite File	Tests	Subsystem Verified	Status
test_feature_extraction.py	6 Tests	Packet parsing, unidirectional flow aggregation, entropy math	PASSED
test_ml_training.py / test_phase3i	12 Tests	HistGradientBoosting, IsolationForest, schema compatibility	PASSED
test_security_hardening.py	8 Tests	Magic bytes, rate limiting, path traversal blocks, bounds checks	PASSED
test_sih26145_requirements.py	8 Tests	DNS DGA, TLS encrypted profiling, incremental PCAP streaming	PASSED
test_multi_capture_evaluation.py	4 Tests	Ground-truth mapping across completely unseen test captures	PASSED
test_health.py / test_inference_api	21 Tests	FastAPI REST contracts, health checks, Supabase connectivity	PASSED

Total Regression Baseline: 59 passed, 0 failed (100% pass rate in 5.57s) | Frontend Build: Built in 164ms (0 errors)

2. Traditional Bidirectional IDS vs. UNIGUARD AI

Architectural Capability	Traditional NIDS (Snort / Suricata)	UNIGUARD AI Platform
Ingestion Model	Requires bidirectional TCP handshakes	100% Unidirectional 5-tuple flow aggregation
Active Probing Footprint	High (Sends TCP Resets, active probe scans)	Zero (100% Passive, zero outbound emission)
Network Attack Surface	Two-way interface vulnerable to exploitation	Unidirectional optical diode boundary (0% TX)
Zero-Day Attack Discovery	Purely signature-matching based	Two-tier ML: 4-Class + Isolation Forest outliers
Forensic Integrity	Standard text syslog records	Chained SHA-256 cryptographic evidence ledger

3. Optional Physical Hardware Specification (Future)

For physical optical isolation deployments, UNIGUARD AI can interface with hardware data diodes (comprising an SFP+ optical transmitter linked via single-strand fiber to an optical receiver). The physical absence of return fiber enforces unidirectional security at the physical layer.

PAGE 10: LIMITATIONS, JUDGE Q&A; & FINAL CONCLUSION

1. Smart India Hackathon Judge Q&A; Reference

Q1: How do you classify flows without return ACK packets?
A: By computing 27 unidirectional metrics (entropy, IAT jitter, TCP flag states) fed into HistGradientBoosting.
Q2: What are the four verified ML classes?
A: BENIGN, DDOS, RECON_SCAN, and C2_BEACONING.
Q3: How do you handle zero-day attacks?
A: Tier 2 Isolation Forest detects multi-dimensional statistical outliers ($score \le 0.0$) independently of labels.
Q4: Can you inspect encrypted TLS traffic?
A: Yes, passively analyzing packet timing jitter and record size distributions without breaking TLS encryption.
Q5: Is data exfiltration an ML class?
A: No. Exfiltration is monitored through statistical volumetric risk scoring.
Q6: Did you build physical hardware?
A: No. UNIGUARD AI is a software-only prototype; physical data diodes represent an optional future deployment layer.

2. Final Project Verification Status Table

Subsystem Area	Status	Technical Verification Summary
UNIGUARD AI Core	IMPLEMENTED	FastAPI microservice + React 19 SPA dashboard
Passive Ingestion Engine	IMPLEMENTED	5-tuple unidirectional flow aggregator (Scapy / DPKT)
27 Canonical Features	IMPLEMENTED	Deterministic Shannon entropy, IAT dynamics, TCP flags
Phase 3I True 4-Class ML	IMPLEMENTED	Frozen HistGradientBoosting (82.24% accuracy on unseen test)
Tier 2 Anomaly Detection	IMPLEMENTED	Isolation Forest outlier scoring (Contamination = 0.02)
DDoS & Recon Detection	VERIFIED	Validated against CSE-CIC-IDS2018 authentic PCAP slices
C2 Beaconing Detection	VERIFIED	Validated against CTU-13 Neris & Ares Botnet PCAP captures
SHA-256 Evidence Ledger	IMPLEMENTED	Cryptographic chaining in Supabase PostgreSQL database
Security Hardening	IMPLEMENTED	Rate limits, 50MB bounds, magic-byte checks, CORS, headers
Physical Data Diode	OPTIONAL / FUTURE	Architectural model ready for physical optical deployment

3. Final Conclusion

UNIGUARD AI (TEAM LIQUID) successfully solves **SIH26145** by delivering an enterprise-grade, mathematically verified, and security-hardened passive threat intelligence platform. By combining unidirectional flow aggregation, 27 deterministic features, two-tier machine learning (HistGradientBoosting + Isolation Forest), and cryptographic SHA-256 evidence chaining, UNIGUARD AI sets a new standard for non-intrusive critical infrastructure defense.